All modules — concept summary

The one-line idea of every section, the results to learn by heart, and where marks get lost

Module I — Crystal structure & X-rays

Crystal, lattice, basis

IN ONE LINE A crystal is one group of atoms (the **basis**) stamped onto every point of an endless regular grid (the **lattice**).

LEARN THIS BY HEART

Simple cubic $N = \frac{8}{8} = 1$ · **Body-centred cubic (BCC)** $N = 1 + \frac{8}{8} = 2$ · **Face-centred cubic (FCC)** $N = \frac{6}{2} + \frac{8}{8} = 4$

WHERE MARKS GET LOST

Two ways this goes wrong in an exam. **Bisecting the wrong thing:** the bisector is perpendicular to the *line joining the two points* and passes through its midpoint — not perpendicular to a cell edge. **Stopping too early or too late:** include a neighbour and its bisector may simply sit outside and do nothing (harmless), but leave out a near one and the cell comes out too big. Safe habit: bisect every neighbour out to about $1.5\times$ the nearest distance, then keep the innermost region.

Symmetry & the 5-fold ban

IN ONE LINE A crystal can only have 1, 2, 3, 4 or 6-fold rotation symmetry — anything else cannot fill space without leaving gaps.

LEARN THIS BY HEART

2-D: **10** point groups, **17** space groups. 3-D: **32** point groups, **230** space groups. Four cheap marks.

Bravais lattices

IN ONE LINE Once only the legal symmetries are allowed, exactly 7 box shapes and 14 distinct lattices survive in 3-D. This is a table to memorise.

WHERE MARKS GET LOST

"Why is there no face-centred *tetragonal* lattice?" is a favourite. Answer: because it is **not new** — a face-centred tetragonal cell can be redrawn as a smaller *body-centred* tetragonal cell (rotate the axes $45°$ and shrink by $\sqrt{2}$). A centring only counts as a separate Bravais lattice if it cannot be re-described as an existing one. That test is what trims the list to 14.

Directions & Miller indices

IN ONE LINE Directions get square brackets $[uvw]$. Planes get round brackets (hkl) and are labelled by the **reciprocals** of where they cut the axes.

LEARN THIS BY HEART

In a cubic crystal, the direction $[hkl]$ is perpendicular to the plane (hkl). This one fact turns most plane questions into simple vector arithmetic.

And a direction $[uvw]$ **lies inside** the plane (hkl) exactly when

$$hu + kv + lw = 0$$

WHERE

(hkl)	Miller indices of the plane
$[uvw]$	indices of the direction
$hu + kv + lw = 0$	the test: if it is zero the direction lies in the plane, or parallel to it

(because "lies in the plane" means "perpendicular to the plane's normal"). No drawing required.

Interplanar spacing

IN ONE LINE Planes with bigger Miller indices sit closer together: $d = a/\sqrt{h^2 + k^2 + l^2}$ for a cube.

Packing & structures

IN ONE LINE Treat atoms as hard spheres. Where they touch fixes a in terms of r — and that gives you the coordination number and the fraction of space filled.

LEARN THIS BY HEART

Structure	Atoms/cell	Touch along	a vs r
Simple cubic	1	cube edge	$a = 2r$
BCC	2	body diagonal	$\sqrt{3}\,a =$
FCC	4	face diagonal	$\sqrt{2}\,a =$
HCP	6	basal edge	$a = 2r$

Diamond & NaCl

IN ONE LINE All three are an FCC lattice with a two-atom basis — only the basis changes.

WHERE MARKS GET LOST

"Why does diamond fill only 34% of space when FCC fills 74%?" Because its bonding is **directional**. Covalent bonds insist on the tetrahedral $CN = 4$ arrangement, and tetrahedra cannot pack tightly. Close packing ($CN = 12$) is what you get when bonding has no preferred direction, as in metals. Structure follows bonding.

X-rays & reciprocal space

IN ONE LINE X-rays reflect off crystal planes only at special angles ($2d \sin\theta = n\lambda$). Describing that in "reciprocal space" turns the condition into a clean geometric picture.

LEARN THIS BY HEART

$$\boxed{2d \sin\theta = n\lambda}$$

WHERE

d	spacing between the reflecting planes
θ	glancing angle: between the X-ray beam and the planes (not to the normal)
n	order of the reflection: 1, 2, 3, ...
λ	wavelength of the X-rays

θ is measured **from the plane**, not from the normal — that is the classic slip. n is the order of the reflection.

WHERE MARKS GET LOST

In the powder method the cone half-angle is 2θ (not θ), and the distance between the two arcs of one cone corresponds to 4θ.

LEARN THIS BY HEART

Reciprocals of the cubic lattices — a near-guaranteed question:

- simple cubic → simple cubic (spacing $2\pi/a$)
- BCC → FCC
- FCC → BCC

Module II — Theory of solids

Free electron theory

IN ONE LINE Treat the metal's loose electrons as a gas of tiny balls. It explains conduction — but gets heat capacity wrong by a factor of 100, because it ignores the Pauli principle.

WHERE MARKS GET LOST

All four failures share one cause: **wrong statistics**. Maxwell–Boltzmann lets every electron soak up thermal energy; the Pauli exclusion principle does not. Sommerfeld fixes the first three by switching to Fermi–Dirac statistics. The fourth needs the *periodic lattice* — and that is band theory.

Fermi energy & friends

IN ONE LINE Trap electrons in a box, fill the levels two at a time from the bottom. The energy of the topmost filled level is the **Fermi energy** E_F, and it depends only on the electron density.

WHERE MARKS GET LOST

Watch the dimension. In **1-D** $D(E) \propto E^{-1/2}$ (falling); in **3-D** $D(E) \propto E^{+1/2}$ (rising). Same particle — only the geometry of counting states changes. Never quote one for the other.

LEARN THIS BY HEART

Typical metal values, worth knowing as sanity checks: $E_F \approx 2$–7 eV, $v_F \approx 10^6$ m/s, $T_F \approx 10^4$–10^5 K.

That last one is the key to Module II: T_F is *enormous* compared with room temperature, so 300 K is effectively "ice cold" for an electron gas — almost nothing is thermally excited.

Density of states

IN ONE LINE $D(E)$ counts how many electron levels sit in each slice of energy. In 3-D it grows as $\sqrt{E}$, which makes the average electron energy $\frac{3}{5}E_F$.

LEARN THIS BY HEART

$$D(E_F) = \frac{3}{2}\frac{N}{E_F}$$

WHERE

$D(E_F)$	density of states exactly at the Fermi energy
N	total number of free electrons
E_F	Fermi energy

LEARN THIS BY HEART

1-D: $D(E) = \dfrac{L}{\pi}\left(\dfrac{2m}{\hbar^2}\right)^{1/2} E^{-1/2}$ — *falls* with energy, diverges at the band bottom.

2-D: $D(E) = \dfrac{Am}{\pi\hbar^2}$ — a constant, independent of E.

3-D: $D(E) = \dfrac{V}{2\pi^2}\left(\dfrac{2m}{\hbar^2}\right)^{3/2} E^{1/2}$ — *rises* as $\sqrt{E}$.

Fermi–Dirac & heat capacity

IN ONE LINE The Fermi function says a level is occupied with probability $f(E)$. Warming a metal only disturbs a thin sliver of width $k_B T$ around E_F — which is why C_v is tiny and grows linearly with T.

Band gaps & Bloch

IN ONE LINE Put the electron in a **repeating** potential instead of a flat one. Bragg reflection at the zone boundary opens gaps, and Bloch's theorem says the wavefunction is a plane wave times a lattice-periodic ripple.

LEARN THIS BY HEART

Gaps open at $k = \pm n\pi/a$, the Brillouin zone boundaries, because that is where Bragg reflection turns travelling waves into standing waves.

Kronig–Penney

IN ONE LINE A solvable repeating potential. One equation, whose left side must stay between -1 and $+1$ — inside that range you get an allowed band, outside it a gap.

LEARN THIS BY HEART

$$\boxed{P\frac{\sin(\alpha a)}{\alpha a} + \cos(\alpha a) = \cos(ka)} \qquad P = \frac{mV_0 ba}{\hbar^2}, \quad \alpha = \sqrt{\frac{2mE}{\hbar^2}}$$

WHERE

P	strength of the barriers $= mV_0 ba/\hbar^2$
α	$\sqrt{2mE}/\hbar$: wave number of the electron between barriers
E	electron energy
a	width of the well between barriers
b	width of a barrier
V_0	height of a barrier
m	electron mass
k	Bloch wave number
$\hbar$	reduced Planck constant

P is the dimensionless *barrier strength* — how tightly the lattice grips the electron.

LEARN THIS BY HEART

The two limits — very commonly asked:

- $P \to \infty$ (huge barriers, isolated atoms): needs $\sin(\alpha a) = 0$, so $\alpha a = n\pi$ and $E_n = \dfrac{n^2 h^2}{8ma^2}$. The bands shrink to sharp atomic levels.
- $P \to 0$ (no barriers, free electrons): gives $\cos(\alpha a) = \cos(ka)$, so $\alpha = k$ and $E = \dfrac{\hbar^2 k^2}{2m}$. The gaps vanish and the free-electron parabola is back.

Also: as energy rises, the $1/\alpha a$ factor shrinks the oscillation, so **bands get wider and gaps get narrower**.

LEARN THIS BY HEART

One band holds exactly $2N$ electrons, where N = number of primitive cells. One valence electron per atom → band half full → metal. Two per atom → band exactly full → insulator, *unless* bands overlap.

Effective mass & classification

IN ONE LINE An electron in a band responds to a field as if it had a different mass $m^* = \hbar^2/(d^2E/dk^2)$ — which can even be negative. A full band carries no current at all.

WHERE MARKS GET LOST

Negative mass looks like nonsense but breaks nothing. m^* is a bookkeeping trick that hides the lattice force: the electron really feels both the applied field *and* the lattice. Near the top of a band the lattice force (Bragg reflection handing momentum back to the crystal) beats the applied force, so the net acceleration opposes the field — and we record that as $m^* < 0$. Total momentum, electron plus lattice, is conserved throughout.

LEARN THIS BY HEART

The one-line test. A solid conducts if and only if there is an *empty* level immediately above an *occupied* one. Partly filled band, or two bands overlapping → metal. Full band with a big gap above it → insulator. Full band with a small gap above it → semiconductor, which is an insulator at $T = 0$ and a conductor when warmed.

Module III — Semiconductor properties

Intrinsic & extrinsic

IN ONE LINE A semiconductor is an insulator with a **small** band gap (about 1 eV). Heat breaks a few bonds and makes electron–hole pairs (intrinsic). Adding a trace of Group V or Group III atoms adds electrons or holes on purpose (extrinsic), and that is how every device is made.

LEARN THIS BY HEART

Properties of a hole: charge $+e$; it moves in the valence band, in the direction of the field; it is not a particle but the absence of an electron in an otherwise full band; it has a **larger effective mass and smaller mobility** than an electron, because it moves in a nearly full band under the applied field *and* the ionic field of the lattice.

LEARN THIS BY HEART

In both types the **law of mass action** holds: $np = n_i^2$ at a given temperature. Doping raises one carrier and lowers the other in proportion. The crystal stays **neutral**: $n + N_a^- = p + N_d^+$.

WHERE MARKS GET LOST

A donor atom does not make a hole when it gives its electron away: what it leaves is a *fixed* positive ion, not a mobile hole. Only broken Si–Si bonds make holes. Likewise an acceptor makes a mobile hole and a fixed negative ion, not a free electron.

Drift, mobility, σ

IN ONE LINE A field gives the carriers a small extra velocity on top of their random motion: $v_d = \mu E$. Electrons and holes both carry current, so $\sigma = e(n\mu_n + p\mu_p)$. In a pure sample σ rises as $e^{-E_g/2kT}$, and $\ln\sigma$ against $1/T$ measures the gap.

LEARN THIS BY HEART

$$v_d = \mu E, \qquad J = \sigma E, \qquad \sigma = e(n\mu_n + p\mu_p)$$

WHERE

v_d	drift velocity
μ	mobility: drift velocity per unit field
E	electric field
J	current density
σ	conductivity
$n,\ p$	electron and hole concentrations
$\mu_n,\ \mu_p$	electron and hole mobilities
e	magnitude of the electron charge

Intrinsic: $n = p = n_i$, so $\sigma_i = en_i(\mu_n + \mu_p)$. Electrons and holes drift opposite ways but, having opposite charges, their currents **add**.

WHERE MARKS GET LOST

The slope of $\ln\sigma$ against $1/T$ is $-E_g/2k$, not $-E_g/k$. Forgetting the 2 halves or doubles your band gap. Mobility data are often given in $\mathrm{cm^2V^{-1}s^{-1}}$: divide by 10^4.

n, p, E_F, n_i

IN ONE LINE Count the states in each band ($D(E)$) times the chance they are filled ($f(E)$). Because E_F sits in the gap, f becomes a simple exponential and the counts come out as $n = N_c e^{-(E_c - E_F)/kT}$ and $p = N_v e^{-(E_F - E_v)/kT}$. Multiply them and E_F drops out: $np = n_i^2$.

LEARN THIS BY HEART

$$n = N_c e^{-(E_c - E_F)/kT}, \qquad p = N_v e^{-(E_F - E_v)/kT}$$

WHERE

$n,\ p$	electron and hole concentrations
$N_c,\ N_v$	effective density of states of the conduction and valence band
$E_c,\ E_v$	band edges
E_F	Fermi level
k	Boltzmann's constant
T	temperature

$$N_c = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2}, \qquad N_v = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2}$$

WHERE

$N_c,\ N_v$	effective density of states of the two bands
$m_n^*,\ m_p^*$	effective masses of the electron and the hole
k	Boltzmann's constant
T	temperature
h	Planck's constant
2	spin degeneracy

These hold for intrinsic *and* doped samples (as long as E_F is a few kT inside the gap). For an intrinsic sample write $E_F = E_i$, the intrinsic level: $n_i = N_c e^{-(E_c - E_i)/kT}$, $p_i = N_v e^{-(E_i - E_v)/kT}$.

LEARN THIS BY HEART

$$np = n_i^2 = N_c N_v e^{-E_g/kT}, \qquad n_i = 2\left(\frac{2\pi kT}{h^2}\right)^{3/2}(m_n^* m_p^*)^{3/4} e^{-E_g/2kT}$$

WHERE

$n,\ p$	electron and hole concentrations
n_i	intrinsic carrier concentration
$N_c,\ N_v$	effective densities of states
E_g	band gap $E_c - E_v$
$m_n^*,\ m_p^*$	effective masses
$k,\ T,\ h$	Boltzmann's constant, temperature, Planck's constant

Valid for intrinsic and extrinsic samples alike (non-degenerate).

WHERE MARKS GET LOST

$n_i \propto e^{-E_g/2kT}$ but $np = n_i^2 \propto e^{-E_g/kT}$. Keep the two exponents apart. And N_c, N_v depend on temperature ($\propto T^{3/2}$): use the value at the temperature you are working at.

Extrinsic E_F & σ

IN ONE LINE In an n-type sample the electrons come from the donors, so E_F moves up, about midway between E_d and E_c. As the sample is heated E_F falls back to mid-gap and the sample behaves intrinsically. Its conductivity is $\sigma \approx N_d e \mu_n$ in the useful range.

LEARN THIS BY HEART

$$E_F = \frac{E_v + E_a}{2} - \frac{kT}{2}\ln\frac{N_a}{N_v}, \qquad p = n_i e^{(E_i - E_F)/kT}, \qquad \sigma_p = ep\mu_p \approx eN_a\mu_p$$

WHERE

E_F	Fermi level of p-type material
E_v	top of the valence band
E_a	acceptor level
N_a	acceptor concentration
N_v	effective density of states of the valence band
p	hole concentration
E_i	intrinsic Fermi level
σ_p	conductivity of p-type material
μ_p	hole mobility
$k,\ T$	Boltzmann's constant, temperature

E_F lies about midway between E_v and E_a, moves up with rising T and finally meets E_i. Not valid at very high T (then n cannot be neglected).

LEARN THIS BY HEART

Summary of the Fermi level: intrinsic, at mid-gap; n-type, about midway between E_d and E_c; p-type, about midway between E_v and E_a. As T rises it approaches the intrinsic level in every case.

Module IV — Magnetic & dielectric properties

Types of magnetism

IN ONE LINE Every solid is magnetic in some way. What kind depends on whether its atoms have **permanent moments** (unpaired electrons) and on how neighbouring moments line up: not at all, randomly, parallel, antiparallel and equal, antiparallel and unequal.

LEARN THIS BY HEART

$$\mathbf{B} = \mu_0(\mathbf{H}+\mathbf{M}) = \mu_0(1+\chi)\mathbf{H} = \mu\mathbf{H} = \mu_0\mu_r\mathbf{H}, \qquad \mu_r = 1+\chi$$

WHERE

$\mathbf{B}$	magnetic flux density (tesla)
μ_0	permeability of free space, $4\pi \times 10^{-7}$ H/m
$\mathbf{H}$	magnetising field
$\mathbf{M}$	magnetisation
χ	susceptibility
μ	permeability of the material
μ_r	relative permeability, $1+\chi$

$\mu_0 = 4\pi \times 10^{-7}$ H/m. B in Wb/m^2 (tesla). In vacuum $M = 0$, $\chi = 0$, $\mu_r = 1$, $B = \mu_0 H$.

LEARN THIS BY HEART

How to tell them apart from data. Sign of χ: negative means diamagnetic. Size: 10^{-5}–10^{-3} means dia- or paramagnetic, 10^{2}–10^{5} means ferro- or ferrimagnetic. Temperature: plot $1/\chi$ against T above the critical point; it is a straight line through the origin (para), cutting the axis at $+T_c$ (ferro) or extrapolating to $-\theta$ (antiferro).

WHERE MARKS GET LOST

Diamagnetism is present in **every** material; it is just usually swamped. A substance is called diamagnetic only when it has no permanent moments.

Dia- & paramagnetism

IN ONE LINE Diamagnetism: a field speeds up or slows down each electron orbit by the Larmor frequency $eB/2m$; the extra current opposes B. Paramagnetism: permanent moments turn towards B against thermal disorder; the Boltzmann average gives the Langevin function and the Curie law $\chi = C/T$.

LEARN THIS BY HEART

$$\chi_{\text{dia}} = -\frac{\mu_0 N Z e^2}{6m}\langle r^2 \rangle$$

WHERE

χ_{dia}	diamagnetic susceptibility
μ_0	permeability of free space
N	atoms per unit volume
Z	electrons per atom
e, m	electron charge and mass
$\langle r^2 \rangle$	mean square radius of the electron orbits

Negative, independent of temperature, and larger for big atoms with many electrons (it grows with $Z\langle r^2 \rangle$). Typical size -10^{-5} to -10^{-6}.

LEARN THIS BY HEART

Curie law $\chi = C/T$, $C = \mu_0 N \mu^2 / 3k$, for $\mu B \ll kT$. Saturation $M_s = N\mu$ when $\mu B \gg kT$ (very low temperature, very strong field).

WHERE MARKS GET LOST

The quantum result uses $\mu^2 = p_{\text{eff}}^2 \mu_B^2 = g^2 J(J+1)\mu_B^2$, not $g^2 J^2 \mu_B^2$. The difference is the "$+1$".

Ferromagnetism

IN ONE LINE Weiss: an internal field proportional to M lines the moments up. It explains spontaneous magnetisation below T_c and the Curie–Weiss law above it. The field is the quantum exchange interaction. A lump of iron can be unmagnetised because it is split into domains; moving and turning them gives the hysteresis loop.

LEARN THIS BY HEART

$$\chi = \frac{C}{T - T_c}, \qquad T_c = \frac{\lambda C}{\mu_0}, \qquad C = \frac{\mu_0 N g^2 J(J+1)\mu_B^2}{3k}$$

WHERE

χ	susceptibility above the Curie point
C	Curie constant
T	temperature
T_c	Curie temperature
λ	molecular-field (Weiss) constant
N	atoms per unit volume
g	Landé factor
J	total angular momentum quantum number
μ_B	Bohr magneton
k	Boltzmann's constant
μ_0	permeability of free space

A strong molecular field (large λ) means a high Curie temperature. Internal field from measured numbers:

$$B_E = \lambda M_s = \frac{3kT_c}{g(J+1)\mu_B} \text{ (quantum)}$$

or $\dfrac{3kT_c}{\mu}$ (classical).

Antiferro & ferri

IN ONE LINE Negative exchange makes neighbours antiparallel. Two equal sublattices cancel (antiferromagnetism: small χ with a cusp at the Néel temperature). Two unequal ones leave a net moment (ferrimagnetism: ferrite, magnetite), which behaves outwardly like a ferromagnet.

LEARN THIS BY HEART

$$\chi = \frac{2C}{T + \theta} \quad (T > T_N)$$

WHERE

χ	susceptibility of an antiferromagnet above T_N
C	Curie constant
T	temperature
θ	constant that sets the intercept of $1/\chi$ at $-\theta$
T_N	Néel temperature

Like Curie–Weiss with the sign of the intercept reversed: $1/\chi$ extrapolates to $-\theta$. In the simplest theory (only A–B interactions) $\theta = T_N$; real crystals have $\theta > T_N$ (MnO: $\theta \approx 610$ K, $T_N = 116$ K).

LEARN THIS BY HEART

Ferrimagnet in one line: antiparallel *and* unequal sublattices. Net moment = difference of the sublattice moments. Outwardly like a ferromagnet (spontaneous M, Curie point, domains, hysteresis), but with a much smaller saturation moment and a very high electrical resistivity.

P, local field, C–M

IN ONE LINE A field pulls the bound charges of a dielectric slightly apart: polarisation P. An atom feels not the average field E but the local field $E + P/3\varepsilon_0$. Putting that into $P = N\alpha E_{\text{loc}}$ gives the Clausius–Mossotti relation between ε_r and α.

LEARN THIS BY HEART

$\mathbf{D} = \varepsilon_0\mathbf{E} + \mathbf{P} = \varepsilon_0\varepsilon_r\mathbf{E}, \qquad \varepsilon_r = 1 + \chi, \qquad P = \varepsilon_0(\varepsilon_r - 1)E$

WHERE

$\mathbf{D}$	electric displacement
$\mathbf{E}$	electric field in the material
$\mathbf{P}$	polarisation
ε_0	permittivity of free space
ε_r	relative permittivity (dielectric constant)
χ	electric susceptibility

P in C/$\mathbf{m}^2$ (it equals the bound surface charge density), $\varepsilon_0 = 8.85 \times 10^{-12}$ F/m.

LEARN THIS BY HEART

Lorentz relation (cubic symmetry):

$$E_{\text{loc}} = E + \frac{P}{3\varepsilon_0}$$

WHERE

E_{loc}	local field acting on one atom
E	average (macroscopic) field
P	polarisation
ε_0	permittivity of free space

The atom feels *more* than the average field, because its polarised neighbours help. In a gas P is tiny and $E_{\text{loc}} \approx E$.

LEARN THIS BY HEART

$$\frac{\varepsilon_r - 1}{\varepsilon_r + 2} = \frac{N\alpha}{3\varepsilon_0} = \frac{1}{3\varepsilon_0}\sum_j N_j\alpha_j$$

WHERE

ε_r	relative permittivity
N	atoms per unit volume
α	polarisability: dipole moment per unit local field
$N_j,\ \alpha_j$	concentration and polarisability of species j (when there are several)
ε_0	permittivity of free space

It connects a measured ε_r to atomic polarisabilities. Exact for cubic crystals and gases; approximate for polar liquids like water.

WHERE MARKS GET LOST

Do not use $\varepsilon_r = 1 + N\alpha/\varepsilon_0$ for a solid: that ignores the local field and is only right for dilute gases (where it agrees with C–M anyway). And ε_r is dimensionless, but α is in F $\mathbf{m}^2$ (C $\mathbf{m}^2$ $\mathbf{V}^{-1}$).

α_e, α_i, α_d

IN ONE LINE Three mechanisms add up: the electron cloud shifts (α_e, every atom), ions shift against each other (α_i, ionic solids), permanent dipoles turn ($\alpha_d = p^2/3kT$, polar molecules). As the frequency rises they drop out one by one: dipolar first, then ionic, leaving the electronic part at optical frequencies.

LEARN THIS BY HEART

$$\alpha_e = 4\pi\varepsilon_0 R^3 \ \text{(static)}, \qquad \alpha_i = \frac{e^2}{\beta}, \qquad \alpha_d = \frac{p^2}{3kT}$$

WHERE

- α_e electronic polarisability
- α_i ionic polarisability
- α_d dipolar polarisability
- ε_0 permittivity of free space
- R atomic radius
- e electron charge
- β restoring-force constant of the ionic bond
- p permanent dipole moment
- k Boltzmann's constant
- T temperature

Only α_d depends on temperature. Plot α against $1/T$: slope $p^2/3k$ gives the dipole moment, intercept gives $\alpha_e + \alpha_i$ (the Debye method).

WHERE MARKS GET LOST

Water's static $\varepsilon_r \approx 80$ but its refractive index is 1.33 ($n^2 = 1.77$): the huge difference is the dipolar part, which is gone at optical frequencies. Never use $\varepsilon_r = n^2$ unless both are at the same frequency.

Ferro- & piezoelectricity

IN ONE LINE A ferroelectric has spontaneous polarisation (the electric twin of ferromagnetism): a P–E hysteresis loop, a Curie point, and $\varepsilon_r \propto 1/(T - T_c)$ above it. A piezoelectric crystal (no centre of symmetry) polarises when squeezed, and strains in a field.

LEARN THIS BY HEART

$$\varepsilon_r = \frac{C}{T - T_c} \quad (T > T_c)$$

WHERE

- ε_r relative permittivity
- C Curie constant (of the electric case)
- T temperature
- T_c ferroelectric Curie temperature

Ferroelectric ↔ ferromagnetic: spontaneous $P \leftrightarrow M$, P–E loop ↔ M–H loop, Curie point and Curie–Weiss law in both. Uses: capacitors (very high ε_r near T_c), non-volatile memory (FeRAM), infrared detectors.

LEARN THIS BY HEART

The ions must be displaced so that the charge distribution loses its symmetry: piezoelectricity needs a crystal with **no centre of inversion** (20 of the 32 point groups). Every ferroelectric is piezoelectric ($BaTiO_3$), but not every piezoelectric is ferroelectric (quartz).